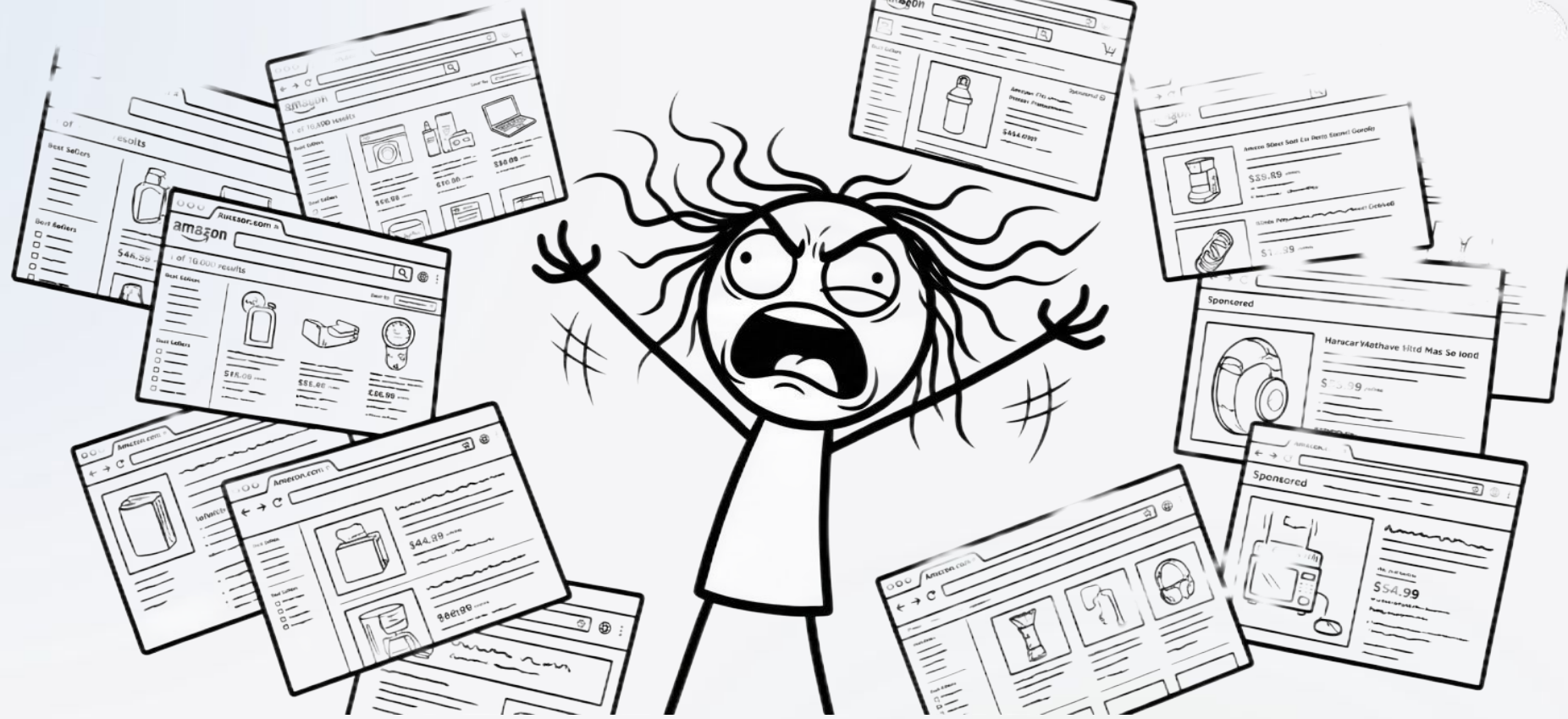




WiseChoice:

An Intelligent Workspace for Online Shopping Decisions



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Introduction and Motivation

Too many challenges ...

- Long, repetitive descriptions (often thousands of words **per page**).
- Large unstructured specification lists (64 comparable attributes).
- Unexplained technical terminology (e.g., Hz, nits, TDP, lumen).

HIGH READING WORKLOAD



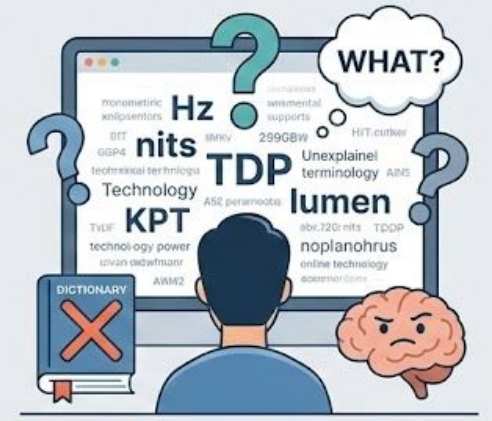
HIGH READING WORKLOAD

HIGH DECISION BURDEN



HIGH DECISION BURDEN

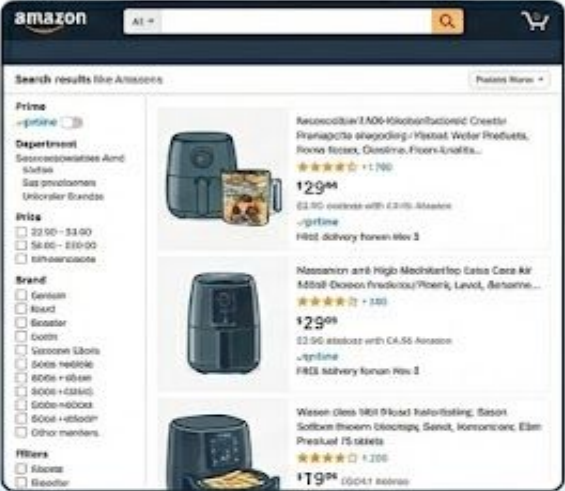
HIGH COMPREHENSION BURDEN



HIGH COMPREHENSION BURDEN

Existing Solutions in Shopping Website and Community *(Amazon as Example)*

AMAZON SEARCH & FILTERS



 Ranks Items

 Decision Factors Hidden



COMPARISON TABLES



 Displays All Parameters

 Importance Unclear

VIDEO / LONG-FORM REVIEWS



 Rich Information

 Time-Consuming & Often Unavailable for Specific Pairs

“

*Can AI reorganize raw e-commerce pages and transform **overwhelming, unstructured** information into a **structured, interpretable, and comparable** decision report, helping users make a Wise Choice?*

”

WiseChoice: A *LLM-Powered Intelligent Workspace*

WiseChoice is acting an
online shopping Copilot,
helping users make a *Wise Choice*.

WiseChoice automatically converts
raw e-commerce pages into
structured and low-cognitive-load
decision reports.

The image displays a screenshot of an Amazon product page for a hair dryer, alongside the WiseChoice interface which provides a structured comparison and analysis of the product.

Amazon Product Page:

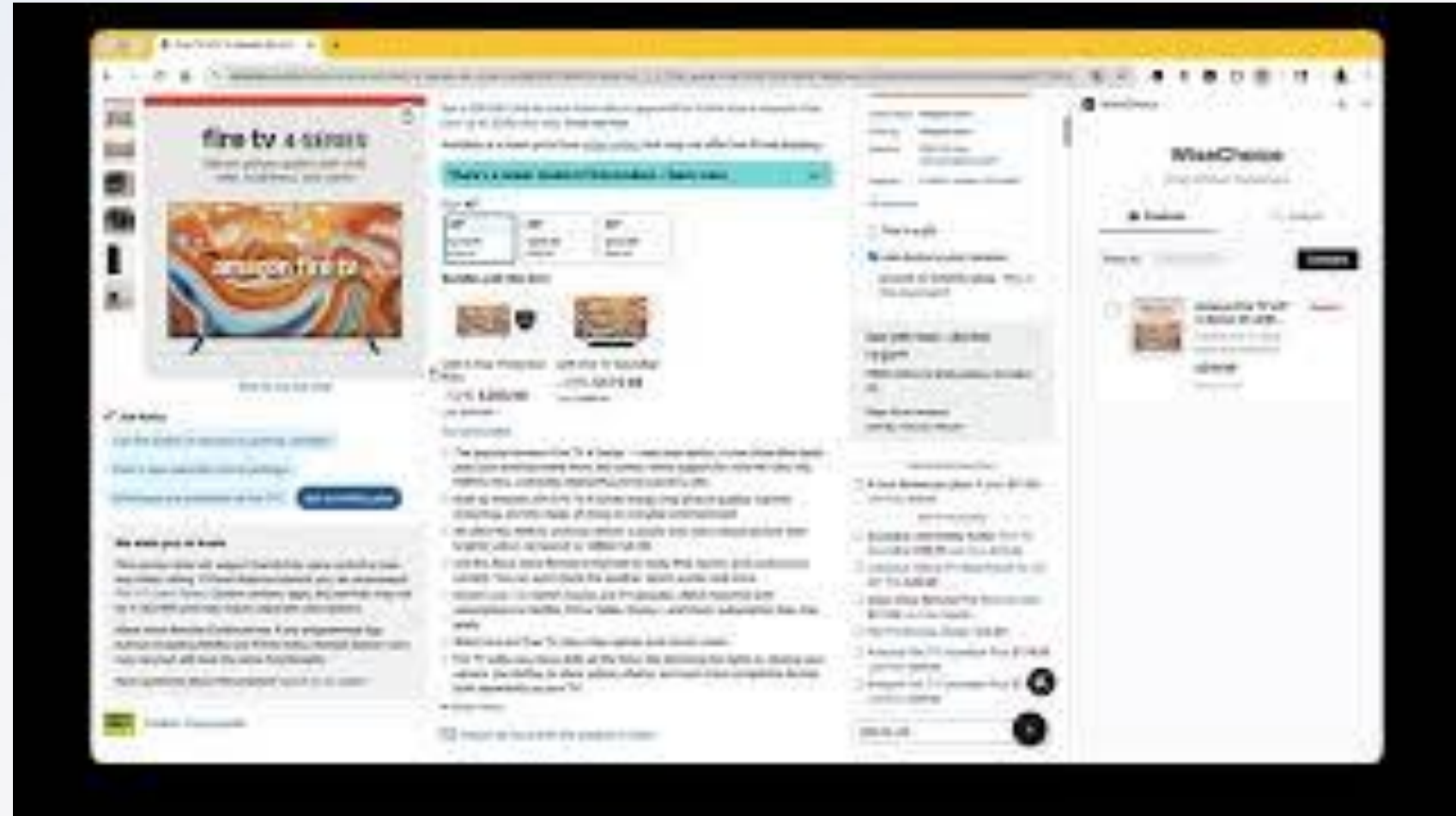
- Product:** Hair Dryer, High-Speed Ionic Quiet Blow Dryer with 160,000 RPM Brushless Motor for Fast Drying, Low Noise Thermo-Control Compact Professional Dryers with Magnetic Diffuser & Nozzle for All Hair Types
- Price:** \$89.99 (\$89.99 / count)
- Prime:** FREE delivery Friday, December 5. Order within 11 hrs 16 mins
- Stock:** In Stock
- Quantity:** 1
- Buttons:** Add to Cart, Buy Now
- Ships from:** Amazon
- Sold by:** ZMZW Direct
- Returns:** FREE refund/replacement until Jan 31, 2026
- Support:** Product support included
- See more:** Add to List
- Brand:** ZMZW
- Color:** Dark Grey
- Material:** Plastic
- Wattage:** 1500
- Power Source:** Corded Electric
- About this item:** Ultra Fast Drying - 160,000 RPM High Speed Motor: Powerful brushless motor delivers up to 36 m/s airflow, drying hair 5x faster than traditional dryers. Perfect for quick styling at home or on the go.

WiseChoice Interface:

- WiseChoice Smart Product Comparison**
- Products:** Hair Dryer, 160000 RPM Professional... (aowoka - ASIN: B0FH4ZB9KL) \$59.99; Hair Dryer, High-Speed Ionic Quiet... (ZMZW Store - ASIN: B0FBWQFF4R) \$89.99
- Analysis:** Fast, ionic travel hair dryer — cheaper, quieter, and with more control vs the pricier alternative
- RECOMMENDATION:** Choose Product 1 (aowoka) — it gives equal or better core tech (160k RPM, ionic output) with more control, clearer noise/weight specs, stronger reviews, and a much lower price.
- What Matters:** Drying performance: prioritize motor speed and airflow (RPM and m/s). RPM (revolutions per minute) describes motor spin; higher generally means stronger airflow and faster drying. Heat & damage control: prefer reliable temperature sensors and multiple heat/speed modes.

Example: Jane Buys a TV

- Jane opens **List Mode** and clears her previous list.
- She adds the two TVs she wants to compare.
- WiseChoice automatically extracts information from both product pages.
- Jane switches to **Report Mode** and views the five-section structured report:
Comparison Title, Recommendation, What Matters, Key Differences, Why This Choice.



WiseChoice transforms the experience
from “scrolling through long pages” to “reading a structured decision report.”

System Design:

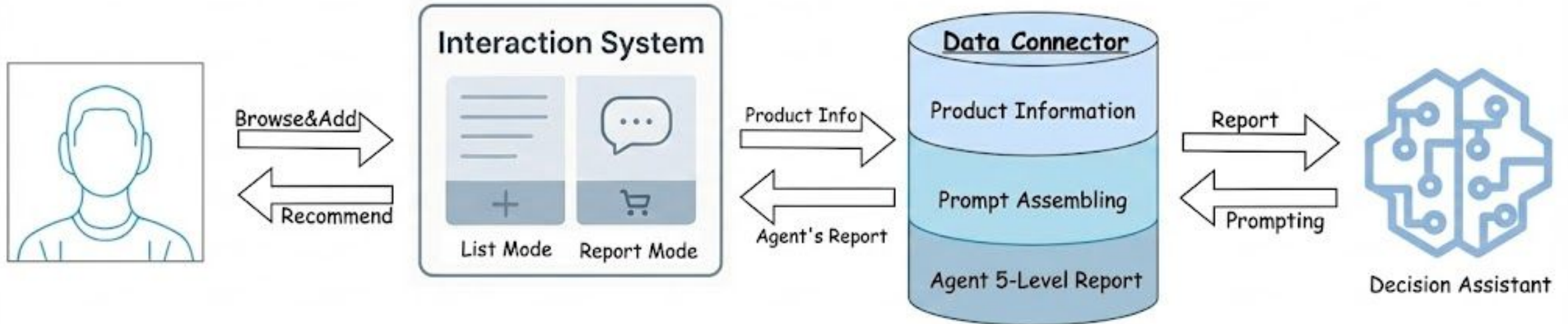
Overall Architecture

User

Dual Mode Side-Workspace

Data Backend

AI Agent



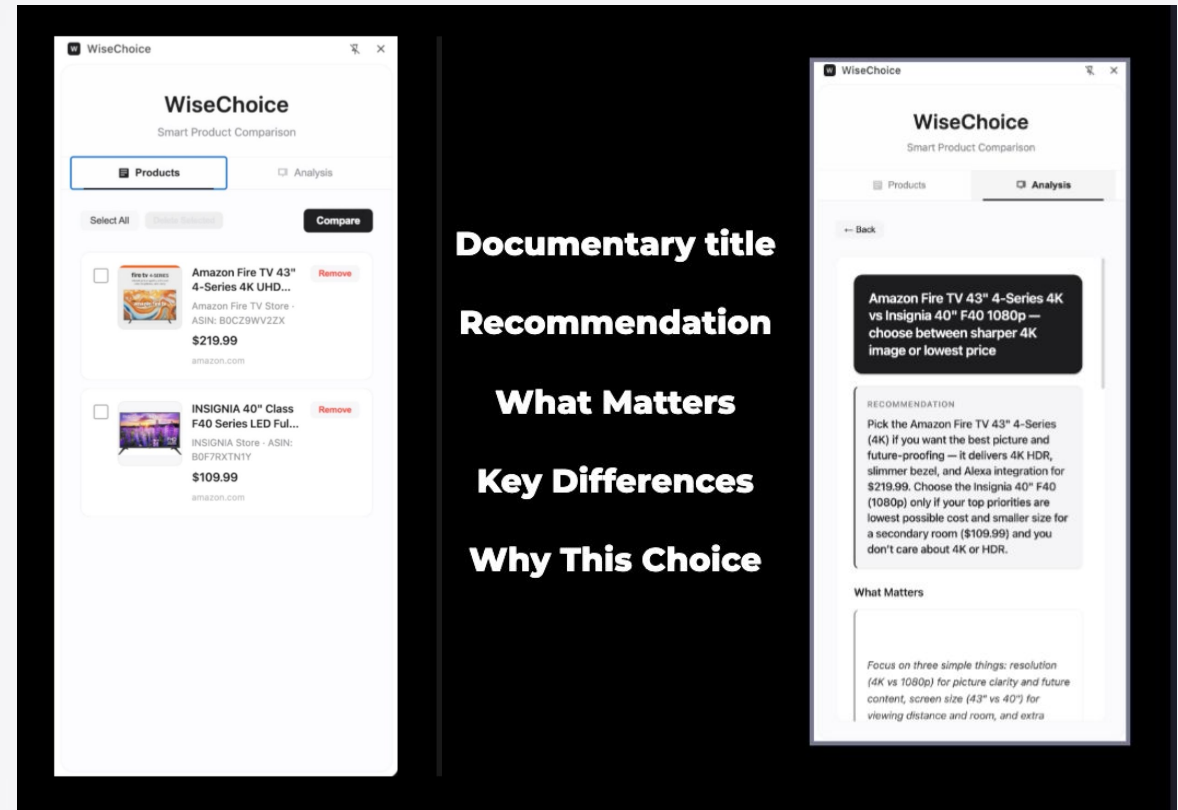
Dual-Mode Workspace *(Interaction System)*

List Mode

Displays current candidate products. Allows adding/removing items. Shows concise information rather than long pages.

Report Mode

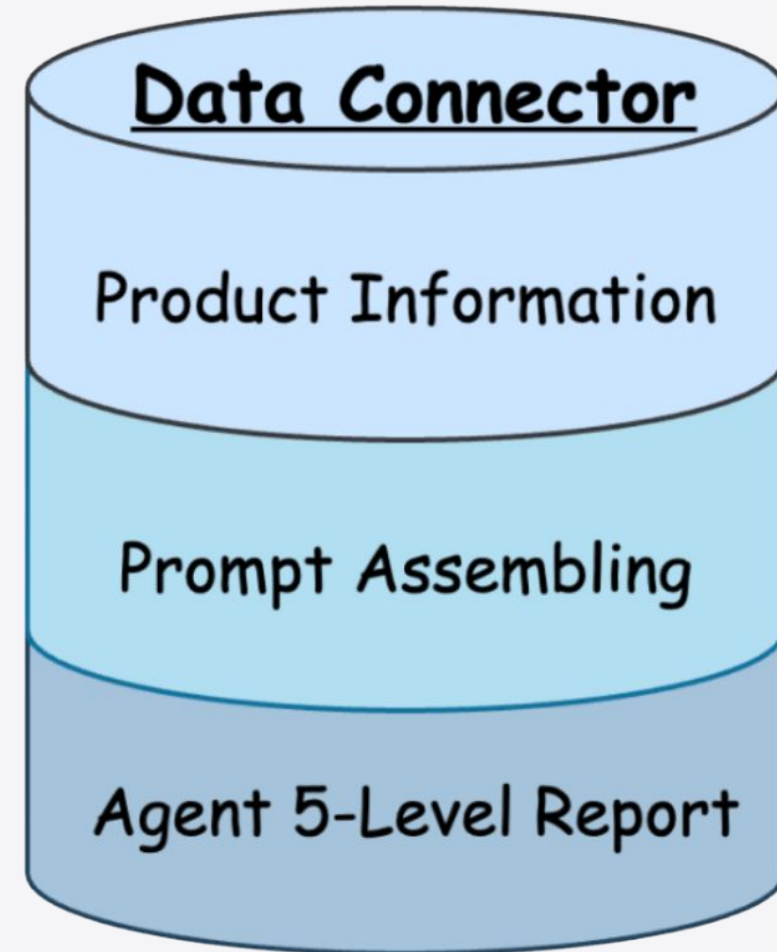
Ensure users no longer need to “jump between pages and manually compare specs,” but instead read structured, explanatory content.



Data Connector

Functions: Data Backend

- Convert Raw **Product Pages** to →
- Cleaned and normalized JSON content →
- Then, **Assembling Prompt** to the Agent →
- Finally, pass the **5-level Report** to workspace



Decision-Making Agent

Responsibilities

As the system's core reasoning engine, the Agent takes the contextual info and prompt from the Data Connector and generates a 5-level report to the Data Connector.

Core Goals

-  Compress essential information
-  Explain unfamiliar concepts
-  Emphasize decision-relevant differences

Output Schema

Comparison Title — defines scenario

Recommendation — one-sentence TL;DR

What Matters — contextual principles

Key Differences — truly matter factors

Why This Choice — detailed reasoning

Evaluation Design: Research Questions

Reading Load

Does WiseChoice reduce reading load?

H1

Decision Efficiency

Does WiseChoice improve decision efficiency?

H2

Terminology Barriers

Does WiseChoice eliminate terminology barriers?

H3

Experimental Setup — Within-subjects design:

- 12 Amazon product pairs
- Raw product pages (Control) vs WiseChoice reports (Treatment)

Measures

Category	Metric	Definition
Reading Load (H1)	Estimated Reading Time	Based on 238 wpm average English silent reading speed (Brysbaert, 2019).
	RR_full	Full Report Information Reduction Ratio (Raw → WiseChoice).
Decision Efficiency (H2)	RR_diff	Key Differences Information Reduction Ratio (Raw → WiseChoice).
	CFCR	Cognitive Factor Compression Rate (attributes in WiseChoice÷ 64).
Terminology (H3)	TBR	Terminology Barrier Rate (proportion of unexplained terms).

Evaluation Results

Significant improvements across all three causal dimensions.

READING LOAD

54.6%

RR_full

Full Report Information Reduction Ratio.

From 4.14m down to 1.80m per decision.

Raw



WiseChoice



DECISION EFFICIENCY

84.5%

RR_diff

Key Differences Information Reduction Ratio.

0.15

CFCR

Cognitive Factor Compression Rate (vs 64 baseline).

TERMINOLOGY BARRIERS

0%

TBR

Terminology Barrier Rate.

The system successfully contextualized every technical term.

Thank You!

Happy to take any questions.

✉ Contact for details: hli52@gmu.edu, <https://haichang.li>